

SLIVEN, Bulgaria

WMO: 156400

Lat: 42.667N

Lon: 26.333E

Elev: 843

StdP: 14.25

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.0	22.3	7.1	8.2	21.2	12.1	10.5	24.5	53.2	30.5	43.8	30.3	10.5	320	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	92.5	73.4	89.5	72.3	87.0	71.0	76.2	87.6	74.3	86.0	72.5	84.2	6.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	126.1	82.0	70.7	116.6	81.1	68.5	107.9	79.9	40.2	87.5	38.4	86.1	36.7	84.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
40.2	31.2	21.6	DB	13.8	97.8	4.2	4.1	10.8	100.8	8.4	103.2	6.0	105.5	3.0	108.5	
			WB	12.3	78.0	4.2	3.0	9.3	80.1	6.8	81.9	4.4	83.5	1.4	85.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	36.0	39.7	46.1	54.4	63.4	71.0	75.7	75.8	67.4	57.1	47.6	38.6
	DBStd	15.43	7.68	8.19	7.28	6.38	5.89	5.40	4.58	4.88	6.07	6.94	7.52	7.60
	HDD50	1437	435	296	164	28	1	0	0	0	0	20	132	361
	HDD65	4268	898	708	586	320	101	11	1	0	46	256	522	819
	CDD50	3684	2	8	43	162	415	630	795	801	521	240	60	7
	CDD65	1041	0	0	0	4	51	191	331	336	117	11	0	0
	CDH74	9022	0	0	1	32	339	1548	3068	3159	809	66	0	0
	CDH80	3225	0	0	0	1	41	477	1198	1278	224	6	0	0
Wind	WSAvg	6.3	5.9	6.6	7.1	5.9	6.0	6.5	7.4	6.6	6.5	6.0	5.1	6.4
Precipitation	PrecAvg	23.1	1.7	1.5	1.6	1.8	2.6	2.5	1.9	1.5	2.2	1.9	1.9	2.1
	PrecMax	41.4	4.0	4.5	3.2	3.5	7.3	7.6	5.5	3.2	6.7	5.5	5.2	5.2
	PrecMin	11.8	0.1	0.0	0.0	0.2	0.8	0.2	0.1	0.1	0.1	0.1	0.0	0.0
	PrecStd	5.8	1.1	1.1	0.9	0.8	1.5	1.7	1.4	1.0	1.9	1.3	1.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.8	63.1	69.6	78.6	84.1	92.4	97.5	96.7	90.0	80.0	70.1	60.9
		MCWB	50.3	53.1	57.2	65.4	67.9	73.9	75.3	73.3	71.8	66.3	60.6	53.9
	2%	DB	53.1	59.1	65.5	72.6	80.5	88.4	92.7	92.7	85.8	75.1	65.2	56.3
		MCWB	47.5	50.5	54.3	61.1	66.2	72.4	73.6	73.4	69.5	63.6	58.3	51.6
	5%	DB	49.7	55.3	61.6	68.8	77.8	85.2	89.3	89.7	82.2	71.2	61.7	52.6
		MCWB	45.6	48.4	52.3	57.8	65.2	70.8	72.1	72.5	67.8	61.6	56.0	49.1
	10%	DB	46.9	51.5	58.0	65.6	75.0	82.4	86.5	87.0	78.7	67.6	58.3	49.6
		MCWB	43.5	46.1	50.1	55.9	63.4	68.9	70.7	71.0	65.9	59.5	53.8	46.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	55.1	58.6	66.9	70.8	78.0	78.5	78.8	74.9	68.7	63.1	57.0
		MCDB	56.3	60.1	67.2	76.5	79.8	86.7	91.1	90.7	85.5	77.0	67.1	59.2
	2%	WB	48.5	51.8	55.6	62.4	68.2	75.0	75.7	76.1	71.6	65.0	59.2	52.4
		MCDB	52.0	57.4	63.3	70.5	78.1	84.0	88.3	88.2	82.1	72.4	63.9	55.2
	5%	WB	46.2	48.9	53.0	59.2	66.2	72.4	73.8	74.2	69.2	62.5	56.8	49.5
		MCDB	49.1	54.3	60.2	66.9	75.5	82.6	86.0	86.6	79.9	69.8	60.9	52.0
	10%	WB	43.8	46.6	50.8	56.7	64.3	70.2	72.0	72.2	67.0	60.3	54.5	46.8
		MCDB	46.6	51.2	57.1	64.1	73.4	80.5	83.9	84.7	76.6	66.7	57.9	49.8
Mean Daily Temperature Range	5% DB	MDBR	10.5	12.4	14.6	16.4	17.4	17.4	18.0	18.6	17.1	14.3	11.7	10.6
		MCDBR	14.2	18.5	21.1	21.5	21.6	21.9	23.3	22.8	22.8	19.9	15.6	13.7
		MCWBR	10.5	12.3	12.8	12.3	11.7	10.6	11.1	11.0	11.4	11.3	10.2	10.3
	5% WB	MCDBR	12.7	17.0	19.8	20.3	20.2	20.2	21.4	21.2	21.5	18.6	13.9	12.3
		MCWBR	10.0	11.9	12.6	12.6	11.6	11.0	11.2	11.2	11.7	11.6	9.6	9.9
Clear-Sky Solar Irradiance	taub	0.345	0.377	0.421	0.475	0.449	0.451	0.457	0.467	0.439	0.401	0.371	0.347	
	taud	2.306	2.221	2.123	1.982	2.133	2.166	2.163	2.130	2.175	2.263	2.293	2.305	
	Ebn at Noon	245	255	257	252	263	262	259	252	249	244	232	232	
	Edh at Noon	28	35	44	55	48	47	46	46	41	34	28	25	
All-Sky Solar Radiation	RadAvg	483	729	1076	1482	1848	2008	2067	1864	1372	890	549	421	
	RadStd	68	89	95	159	143	155	144	97	111	93	63	58	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.92	N/A	N/A	N/A	+2.20	+2.81	-160	-229	N/A	N/A
(47)	Regional (30 neighbors)	+0.76	N/A	N/A	N/A	+1.03	+1.04	N/A	-205	N/A	N/A

Nomenclature: See separate page